MIL-P-45107A(Ord)
9 November 1959
SUPERSEDING
MIL-P-45107(Ord)
19 June 1958

MILITARY SPECIFICATION

PERISCOPE, M29

1. SCOPE

1.1 This specification covers one type of periscope known as Periscope, M29 for sighting and observing with the 120MM Gun Tank, M103A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

O-E-760	Ethyl Alcohol (Ethanol)
UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-P-313	Paper, Lens Tissue
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
CCC-C-271	Cheesecloth, Bleached and Unbleached
JJJ-C-561	Cotton; Sterile and Nonsterile
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-601	Boxes; Wood, Cleated Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
PPP-B-636	Boxes; Fiber
PPP-C-843	Cushioning Materials, Cellulosic
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant

Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-B-130	Barrier Material, Paper, Noncorrosive
MIL-L-6085	Lubricating Oil; Aircraft Instrument, Low Volatility
MIL-C-7769	Cushioning Material, Bound
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping

FSC 1240

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Ordnance Corps

F8289171	Periscope, M29
SQAPL-8289171	List of Supplementary Quality Assurance Provisions for Periscope, M29

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one periscope representative of production shall be made available to the Government to determine if the periscope meets all the requirements of this specification.

3.2 Materials.- Material shall be in accordance with applicable drawings and specifications forming a part of this specification.

3.3 Construction.- The periscope shall be manufactured in accordance with Drawing F8289171, Periscope, M29 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product prior to final acceptance by the Government.

3.4 General specifications.-The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-F-13926 and MIL-O-13830.

3.4.1 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.4.2 MIL-O-13830

(a) Cleanliness

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3.5 Environmental.

3.5.1 Storage temperature.- The periscope shall show no evidence of failure such as cement separation, glass breakage or other physical failure, and shall meet all requirements of this specification at standard ambient temperature (60 degrees Fahrenheit to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.5.2 Operating temperatures.- The periscope shall meet the applicable requirements of 3.9.5 and 3.10.5 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees Fahrenheit and minus 40 degrees Fahrenheit.

3.5.3 Vibration.- There shall be no visible damage or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds with the input coupling immobilized and the complete periscope mounted in its normal operating position. Subsequent to vibration the periscope shall meet all of the following requirements of this specification.

3.6 Cleanliness.- The periscope shall meet the cleanliness requirements of Specification MIL-O-13830, and there shall be no more than 2 particles of dirt visible in the reticle viewing area of the 8 power system or in the reticle viewing area of the unity power system, and each particle shall not be in excess of an area equivalent to a size ten dig. There shall be no dirt on other surfaces which interferes with observation or adversely affects image quality.

3.7 Parallax.- With the parallax adjustment centered at the mid-point of the objective travel, parallax between the images of two targets, one from infinity through the reticle viewing system and one through the target viewing system, observed between 500 to 1500 yards, shall not exceed 0.2 mil.

3.8 Sealing.- The periscope head and body, when tested separately, shall show no evidence of leakage in excess of 0.05 psi when subjected to an internal pressure of 5 psi for a period of five minutes after having been exposed to the requirements of 3.5. The interior atmosphere of the accepted head and body shall consist of nitrogen having a dew-point at least as low as minus 25 degrees Fahrenheit at atmospheric pressure.

3.9 Head assembly.* - With Datum Plane "X" in a vertical plane within 10 seconds, Datum Plane "Y" in a horizontal plane within 10 seconds, and Side "Y" of the Input Coupling (F8289173) positioned 67 degrees and 30 minutes with the horizontal, the head assembly shall meet the following requirements.

*NOTE: The following references apply throughout this specification

Head assembly

Datum Plane "X", Drg. F8289176 (locating pads)
Datum Plane "Y", Drg. F8289176 (horizontal stops)
Datum Plane "Z", Drg. F8289173 (body mounting surface)
Area "C", Drg. F8289260 (8x aperature)

Body Assembly

Reference line "Y", Drg. F8289261 (intersection of planes)

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3.9.1 Collimation.- With an incident line of sight perpendicular to Datum Plane "X", the following requirements shall be met:

3.9.1.1 Eight power system.- The reflected line of sight when measured in Area "C" shall be perpendicular to Datum Plane "Z" within 0.2 mil laterally and 0.2 mil in elevation.

3.9.1.2 Unity power system.- The reflected line of sight in the unity power viewing area shall be laterally perpendicular to Datum Plane "X" within 15 minutes and shall be in a plane depressed 20 degrees to the horizontal within 15 minutes.

3.9.2 Plumb travel.- The line of sight shall not deviate from a plumb line by more than 0.25 mil when elevated 22 degrees and depressed 18 degrees by means of the coupling, starting with the line of sight coincident with the plumb line at the horizontal position.

3.9.3 Elevation accuracy.- The change in angle at the input coupling shall cause the same angular change in the line of sight within 0.2 mil, from 18 degrees depression to 22 degrees elevation.

3.9.3.1 Backlash.- Backlash shall not exceed 0.05 mil.

3.9.4 Coupling travel.- Rotation of elevation input coupling shall be limited by contact with the travel stop pin only.

CAUTION: At no time shall the elevation input coupling be allowed to strike against the travel stop pin.

3.9.5 Torque.- The running torque required to rotate the coupling through maximum elevation and depression shall not exceed seven pound inches at standard ambient temperature (60 degrees Fahrenheit to 90 degrees Fahrenheit) and the torque shall not change from the value obtained at room temperature by more than three pound inches when subjected to the temperatures specified in 3.5.2.

3.9.6 Definition.- The resolving power on the optical axis shall be six seconds of arc or better when viewing through the target viewing system (Area "C") and 45 seconds or arc or better for the remaining area.

3.10 Body assembly (see Note, page 3)

3.10.1 Collimation.- An image projected into the target viewing system normal to the mounting surface shall coincide with a second image projected into the reticle viewing system parallel to Reference Line "Y" within 0.2 mil as measured in the target viewing system.

3.10.2 Image tilt.- With the mounting surface vertical and Reference Line "Y" horizontal, the two images of a plumb line formed by the target viewing system and the reticle viewing system respectively, shall be vertical within 1 degree and parallel within 1 degree.

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3.10.3 Definition.- The resolving power on the optical axis shall be 6 seconds or better when viewed through the target viewing system.

3.10.4 Eyepiece focus.- The diopter scale shall indicate zero within 1/4 diopter when the image of the reticle viewing system is brought into sharp focus.

3.10.4.1 Diopter movement.- The eyepiece shall be capable of movement throughout a range of plus or minus 4 diopters.

3.10.4.2 Backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter.

3.10.5 Torque.- The running torque required to rotate the diopter adjustment shall not exceed five pound inches at standard ambient temperature (60 degrees Fahrenheit to 90 degrees Fahrenheit) and the torque shall not change from the value obtained at room temperature by more than three pound inches when subjected to the temperatures specified in 3.5.2.

3.11 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation or production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

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4.3 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed on Supplementary Quality Assurance Provisions List (SQAPL) 8289171.

4.3.1 The contents of SQAP shall represent the minimum inspection requirements, and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

4.3.2 Contractor's inspection for all components not covered by SQAP shall be in accordance with the inspection procedures of the written inspection plan required by 4.2 and 4.2.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements that have a bearing on the functional accuracy and performance of the Periscope M29.

4.4 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.5 Production model.

4.5.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.5.2 These tests shall be conducted at the contractor's facilities using test equipment designed or procured for use on the remaining items.

4.5.3 The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

4.6 Inspection provisions.

4.6.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.6.2 Examination.- The classification of defects in tables I, II and III shall constitute the minimum inspection requirements to be performed prior to acceptance.

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4.6.2.1 All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

The AQL value is expressed in percent defective

MAJOR: AQL 4.0%	Req't Par.	QA Par.	Test Method
1. Physical failure (post storage)	3.5.1	4.9.1.1	Tactile-Visual
2. Physical failure (post vibration)	3.5.3	4.9.1.3	Device Special Testing (DST)
3. Excessive coupling (+150°F, -40°F)	3.5.2	4.9.1.2	DST
4. Excessive diopter torque (+150°F, -40°F)	3.5.2	4.9.1.2	DST

TABLE II - CLASSIFIED OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105
AQL values are expressed in percent defective

MAJOR: AQL 1.5%	Req't Par.	QA Par.	Test Method
1. Cleanliness	3.6	4.9.1.4	Visual
2. Parallax	3.7	4.9.1.4	DST
3. Sealing	3.8	4.9.2.2	DST
4. Identification marking	3.11	4.9.5	Visual
Head Assembly			
4. Collimation	3.9.1	4.9.3.1	DST
5. Plumb travel	3.9.2	4.9.3.2	DST
6. Elevation accuracy	3.9.3	4.9.3.3	DST
7. Backlash	3.9.3.1	4.9.3.4	DST
8. Coupling travel	3.9.4	4.9.3.5	DST
9. Coupling torque	3.9.5	4.9.3.6	DST
10. Definition	3.9.6	4.9.3.7	DST

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MAJOR: AQL 1.5% (Cont'd)	Req't Par.	QA Par.	Test Method
Body Assembly			
11. Collimation	3.10.1	4.9.4.1	DST
12. Image tilt	3.10.2	4.9.4.2	DST
13. Definition	3.10.3	4.9.4.3	DST
14. Eyepiece focus	3.10.4	4.9.4.4	DST
15. Diopter movement	3.10.4.1	4.9.4.5	DST
16. Backlash	3.10.4.2	4.9.4.6	DST
17. Torque	3.10.5	4.9.4.7	DST

TABLE III - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL values are expressed in percent defective

MAJOR: AQL 2.5%	Req't Par.	Test Method
1. Markings illegible, missing on bag, shipping containers (Levels A,B,C)	5.5 5.5.1 5.5.2 5.5.3	Visual MIL-STD-129 MIL-P-116
2. Water-vaporproof bag damaged or improperly sealed (Level A)	5.3.1.4.4	Visual MIL-B-117 MIL-P-116
3. Improper cleaning (Level A,C)	5.3.1.1 5.3.2.1	Visual MIL-P-116
4. Improper drying (Level A,C)	5.3.1.2 5.3.2.1	Visual MIL-P-116
MINOR: AQL 4.0%		
43. Preservative improperly applied or missing (Level A)	5.3.1.3	Visual MIL-L-6085 MIL-P-116
44. Unit packaging (Level A,C) Method IA-8, III	5.3.1.4 5.3.2.2	Visual MIL-P-116
45. Wrapping preservative coated surfaces omitted (Level A), incorrect barrier material	5.3.1.4.1	Visual MIL-B-121
46. Lens tissue missing, cushioning material incorrect (Level A)	5.3.1.4.2	Visual PPP-C-843
47. Wrapping improperly applied, barrier material not as specified (Level A)	5.3.1.4.3	Visual MIL-B-130

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MINOR: AQL 4.0% (Cont'd)	Req't Par.	Test Method
48. Cushioning and immobilizing; incorrect fiberboard box (Level A,B)	5.3.1.4.5 5.3.1.4.6	Visual PPP-B-636 MIL-C-7769
49. Packing of unit packages not immobilized, shipping container not conforming to specifications (Levels A,B,C)	5.4.1 5.4.2 5.4.3	Tactile-Visual PPP-B-601 PPP-B-621 PPP-B-585 PPP-B-636
50. Strapping (Level A)	5.4.4	Visual
51. Gross weight exceeded (Levels A)	5.4.3.1	Standard Measuring Equipment (SME)
52. Workmanship	5.7	MIL-P-116 Visual

TABLE IV - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

The AQL value is expressed in percent defective

MAJOR: AQL 4.0%	Req't Par.	Test Method
1. Heat seal (Level A)	5.3.1.4.4	MIL-P-116
2. Quick leak (Level A)	5.3.1.4.4	MIL-P-116

4.7.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.8 Inspection equipment.

4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment and to use it for Government verification inspection or acceptance inspection.

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4.8.2 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.3 Accuracy requirements for device, special testing.

4.8.3.1 Device to test vibration requirements (complete periscope).- Complete periscope is defined as that of the complete head and body assembly as depicted on Drawing F8289171). The device shall consist of a vibrating machine which will produce a vertical motion at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) equipped with a means capable of holding the complete periscope in the following manner. The periscope shall be held in the normal operating position with the window in the uppermost position, as depicted on Drawing F8289171, with Arm Assembly C8298397 completely immobilized and the base of the medium secured firmly to the vibrating machine.

4.8.3.2 Device to test head assembly, collimation, eight power system, unity power system, plumb travel, elevation accuracy, definition and backlash.

4.8.3.2.1 Head assembly.- The device shall permit mounting the periscope head assembly in a vertical position with the window in the uppermost position, as depicted on Drawing F8289176, so that the four pads of datum plane "X", Drawing F8289176, are in complete contact with the vertical mating surface "X" of the device. The vertical mating surface "X" of the device shall be vertical within ten seconds. The horizontal mating surface "Y" of the device (Datum Plane "Y"), Drawing F8289176) shall be horizontal within ten seconds.

4.8.3.2.2 Collimation.- Mating surface "Y" of the device input coupling (side "Y" of Input Coupling F8289173) shall be positioned 22 degrees 30 minutes plus or minus 10 seconds with respect to mating surface "X" of the device, and arrested in this position in order to satisfy collimation requirements and the zero line of sight.

4.8.3.2.2.1 Eight (8X power system).- The medium for testing the reflected line of sight measured in area "C", Drawing F8289260, with respect to Datum Plane "Z", shall be accurate within ten seconds throughout the eight power system side of the device.

4.8.3.2.2.2 Unit power system.- The medium used for testing the reflected line of sight in the unity power viewing area with respect to Datum Plane "X" and the depressed plane of 20 degrees, shall be accurate within two minutes throughout the unity power system side of the device.

4.8.3.2.3 Plumb travel.- The medium used to test plumb travel shall be perpendicular within ten seconds with respect to the mating surface, Datum Plane "Y" of the device.

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4.8.3.2.4 Elevation accuracy.- The device input coupling may compensate, both in the vertical and horizontal plane, but the mating surface "Y" shall remain stable within ten seconds in the arrested zero line of sight position; also throughout the entire excursion of rotation for testing angular accuracy.

4.8.3.2.5 Definition.- The medium used for testing the resolving power and the methods and accuracies governing the resolution test shall be in strict accordance with requirements contained in Specification MIL-O-13830.

4.8.3.2.6 Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of device accuracy shall be furnished, and will become a part of the device.

4.8.3.2.7 Device to test torque requirements (head assembly).- The device shall have a minimum range of from zero to ten pound inches, the accuracy of which shall be plus or minus 1/4 pound inches throughout the entire zone of excursion, in both the clockwise and counterclockwise directions. The device shall be equipped with an adapter which will fit snugly over the tapered Dovetail Arm Assembly, Drawing C8298397.

4.8.3.3 Device to test body assembly, collimation, image tilt, definition and eyepiece focus.

4.8.3.3.1 Collimation.- The image projected into the target viewing system shall be normal to the device mounting surfaces within ten seconds in both the vertical and horizontal position, and shall be in coincidence with the second image projected into the reticle viewing system within ten seconds. The respective images shall be parallel to the device mating surface of reference line "Y", Drawing F8289261, within ten seconds.

4.8.3.3.2 Image tilt.- The plumb lines used for testing image tilt shall be perpendicular respectively within ten seconds to mating surfaces reference line "Y" of the device.

4.8.3.3.3 Definition.- The medium used for testing the resolving power and the methods and accuracies governing the resolution test shall be in strict accordance with requirements contained in Specification MIL-O-13830.

4.8.3.3.4 Eyepiece focus.- The medium or dioptometer used to test eyepiece focus shall meet the following accuracy requirements: Any diopter setting of the dioptometer shall agree with the corresponding scale reading within plus or minus 1/4 diopter for a corresponding motion of the eyepiece.

4.8.3.3.5 Device to test torque requirements (body assembly).- The device shall have a minimum range of from zero to ten pound inches, the accuracy of which shall be plus or minus 1/4 pound-inches throughout the entire zone of excursion in both the clockwise and counterclockwise directions. The device shall be equipped with an adapter adjustment of the body assembly in both the clockwise and counterclockwise directions.

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4.9 Test methods and procedures.

4.9.1 Environmental tests.

4.9.1.1 Storage temperature.- The periscope shall be placed in a device, special testing, and exposed to the temperature requirements of 3.5.1 after which it shall meet the requirements of this specification.

4.9.1.2 Operating temperature.- The periscope shall be exposed to the temperature requirements of 3.5.2, and while stabilized at the temperatures listed, shall meet the requirements of 3.9.5 and 3.10.5.

4.9.1.3 Vibration.- The periscope shall be mounted upon a device, special testing, with the input coupling immobilized and vibrated in compliance with the requirements of 3.5.3.

4.9.1.4 Cleanliness.- Subsequent to vibration, the periscope shall be examined thoroughly for compliance with 3.6.

4.9.2 Performance.

4.9.2.1 Parallax.- With the periscope mounted upon a device, special testing, and the required adjustments made, measurable parallax through the target viewing system and through the reticle viewing system shall not exceed the tolerance requirements of 3.7.

4.9.2.2 Sealing.- The periscope, head and body, shall be subjected to the internal pressure test outlined in 3.8, and shall comply with the requirements of 3.9.1, 3.9.1.1, and 3.9.1.2.

4.9.3 Head assembly.- The head assembly shall be mounted upon a device, special testing, in compliance with the requirements of 3.9, in preparation for the following tests.

4.9.3.1 Collimation.- Collimation shall be in compliance with the requirements of 3.9.1, 3.9.1.1, and 3.9.1.2.

4.9.3.2 Plumb travel.- With the periscope mounted upon a device, special testing, the line of sight shall track a plumb line through elevation and depression as directed, within the requirements of 3.9.2.

4.9.3.3 Elevation accuracy.- The change in angle at the input coupling and the line of sight shall comply with the requirements of 3.9.3.

4.9.3.4 Backlash.- In the preceding function (4.9.3.3) shall not exceed the tolerance listed in 3.9.3.1.

4.9.3.5 Coupling travel.- Rotational limits of the input coupling shall comply with the requirements of 3.9.4.

4.9.3.6 Torque.- The running torque required to rotate the coupling shall be measured with a device, special testing, as prescribed in 3.9.5, and shall not exceed the values listed therein.

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4.9.3.7 Definition.- The resolving power of the optical system shall comply with the requirements of 3.9.6.

4.9.4 Body assembly.- The body assembly shall be mounted upon a device, special testing, in compliance with the requirements of 3.10, in preparation for the following tests.

4.9.4.1 Collimation.- Collimation shall be in compliance with the requirements of 3.10.1.

4.9.4.2 Image tilt.- Verticality and parallelism of the image shall comply with the requirements of 3.10.2.

4.9.4.3 Definition.- The resolving power of the optical axis shall comply with the requirements of 3.10.3.

4.9.4.4 Eyepiece focus.- The zero setting of the diopter scale shall comply with the requirements of 3.10.4 when sharp focus is obtained.

4.9.4.5 Diopter movement.- The range of movement of the eyepiece shall comply with the requirements of 3.10.4.1.

4.9.4.6 Backlash.- The diopter movement backlash shall be checked for compliance with 3.10.4.2.

4.9.4.7 Torque.- The running torque required to rotate the diopter movement shall be measured with a device, special testing, and shall comply with the requirements of 3.10.5.

4.9.5 Identification marking.- Markings shall be visually inspected for conformance to Standard MIL-STD-130.

4.9.6 Final acceptance.- Final acceptance of a lot of periscopes shall be withheld by the Government inspector until the sample periscopes, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

4.10 Packaging, packing and marking.

4.10.1 Preproduction pack.- A preproduction pack shall be inspected by the contractor or his agent and witnessed by the Government inspector to determine conformance with the requirements of this specification for the packaging and packing levels specified. The rough handling test shall be performed on the preproduction pack.

4.10.1.1 Preproduction pack failure.- Should the preproduction pack fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be completely inspected by the contractor and witnessed by the Government inspector.

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4.10.2 Sample packs for heat seal, quick leak.- Sample packs for heat seal and quick leak tests shall be selected from acceptable packs of the samples that have met the minimum inspection requirements listed in Table IV.

4.10.3 Cleanliness, rough handling, preservative compounds application, heat seal and quick leak tests.- Cleanliness, rough handling, preservative compounds application, heat seal and quick leak tests shall be conducted as specified in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

5.2 Size of container.- Bags, interior containers and exterior containers shall be of a minimum size to adequately contain the item.

5.3 Preservation and packaging.

5.3.1 Level A.

5.3.1.1 Cleaning.

5.3.1.1.1 Optics.

5.3.1.1.1.1 The following materials and equipment shall be used in the cleaning of the exposed glass surfaces of the optics:

Materials and equipment	Specification
Alcohol, ethyl	Grade II of Specification O-E-760
Bottle, glass, 4 ounce	----------
Brush, camel's hair	----------
Cheesecloth	Specification CCC-C-271
Cloth, batiste	----------
Cotton, absorbent	Grade A, class 1, of Specification JJJ-C-561
Paper, lens tissue	Specification UU-P-313
Sticks, orange or equal	----------
Syringe, rubber	----------

5.3.1.1.1.2 Cleaning the exposed glass surfaces shall be accomplished in the following manner:

(a) Loose particles of dust shall be removed from the surfaces of the optic with a camel's hair brush.

(b) The surface to be cleaned shall be wiped with a circular motion using cotton, lens paper or freshly laundered cheesecloth which is saturated with ethyl alcohol. The surface shall be dried with clean cotton, lens paper or freshly laundered cheesecloth.

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(c) A swab shall be made by wrapping one piece of lens tissue paper around the end of an orange stick or equal. One or two drops of alcohol shall be added on the tip of the swab. The swabbing operation shall begin at the center of the polished surface. A light, downward pressure shall be exerted on the swab and with a circular motion gradually increase the radius of area being cleaned until the entire surface has been swabbed. This operation shall be repeated until no trace of dirt, lint or smears are perceptible.

(d) A rubber syringe shall be used as a bellows to remove any residual dust or lint that may remain on the cleaned surface.

(e) Immediately after cleaning, the optics shall be covered or wrapped with lens tissue paper conforming to Specification MIL-P-13988. The lens tissue paper shall be secured in place with pressure-sensitive tape conforming to type I of Specification UU-T-106.

5.3.1.1.2 Other surfaces.- Cleaning Process C-1 of Specification MIL-P-116 shall be used to clean all other surfaces of the periscope.

5.3.1.2 Drying.- Immediately after cleaning, the item shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the item.

5.3.1.3 Preservation application.- The exposed metal surfaces of the coupling wedge shall be coated with preservative conforming to Specification MIL-L-6085 (P-17 of Specification MIL-P-116). The preservative shall be applied by the brushing method. All excess preservative shall be removed.

5.3.1.4 Unit packaging.- Each periscope shall be packaged as specified herein and as shown in Figures 1, 2 and 3, which is in conformance with Method IA-8 of Specification MIL-P-116.

5.3.1.4.1 Wrapping preservative coated surfaces.- The preservative coated surfaces of the wedge shall be wrapped with greaseproofed barrier material conforming to type II, grade A, class 2 of Specification MIL-B-121. The barrier material shall be secured in place with pressure sensitive tape conforming to the tape specified in 5.3.1.1.1.1(e).

5.3.1.4.2 Cushioning optics.- The tissue covered optics shall be cushioned with cellulosic cushioned with cellulosic cushioning material at least $\frac{1}{2}$ inch thick conforming to type III, class A or B of Specification PPP-C-843. The cushioning material shall be secured in place with the pressure sensitive tape specified in 5.3.1.1.1.1(e).

5.3.1.4.3 Wrapping the periscope.- The periscope shall be wrapped with barrier material conforming to type I of Specification MIL-B-130 and overwrapped with paperboard cushioning conforming to type I, class 1 of Specification PPP-P-291. The wrap shall be secured in place with gummed tape conforming to class 2 of Specification UU-T-111.

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5.3.1.4.4 Water-vaporproof bag.- The wrapped periscope shall be placed in a bag conforming to type I, class e of Specification MIL-B-117. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

5.3.1.4.5 Fiberboard box.- Each bagged periscope shall be cushioned and immobilized (specified in 5.3.1.4.6) in a fiberboard box conforming to class 2 or 3, any compliance symbol and any style, except FTC, TS, OPF and FPF of Specification PPP-B-636.

5.3.1.4.6 Cushioning, immobilizing and closing the fiberboard box.- The interior surfaces (sides, ends, top and bottom) of the fiberboard box shall be lined with 2 inch thick bound cushioning material conforming to type III or IV of Specification MIL-C-7769. The bagged sight shall fit snugly in the bound cushioning material. The fiberboard box shall be closed with pressure sensitive tape conforming to Specification PPP-T-76 in accordance with the appendix of Specification PPP-B-636.

5.3.2 Level C.

5.3.2.1 Cleaning and drying.- Cleaning and drying shall be as specified in 5.3.1.1 and 5.3.1.2.

5.3.2.2 Unit packaging.-Each periscope shall be packaged as described in 5.3.1.4.1 to 5.3.1.4.6 inclusive, except as follows, which is in conformance with Method III of Specification MIL-P-116.

(a) Since preservative is not required, the greaseproofed barrier material, specified in 5.3.1.4.1 shall be omitted.

(b) The water-vaporproof bag, specified in 5.3.1.4.4 shall be omitted.

(c) A fiberboard box conforming to type I or II, class 1, any style, except FTC, TS, OPF and FPF, of Specification PPP-B-636 shall be used in lieu of the fiberboard box specified in 5.3.1.4.5. The box shall be closed as specified in the appendix of Specification PPP-B-636 in lieu of the method specified in 5.3.1.4.6.

5.4 Packing.

5.4.1 Level A.- Six unit packaged periscopes shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - type II or III, class 2 plywood

PPP-B-621 - Class 2 (oversea) style 2 or 4

5.4.2 Level B.- Six unit packaged periscopes shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style, except style 1, class 1, construct for type 3 load

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PPP-B-601 - Any style under domestic type

PPP-B-621 - Class 1 (domestic), style 2 or 4

5.4.3 Level C.- A number of unit packaged periscopes shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style except style 1, class 1, construct for type 3 load

PPP-B-621 - Class 1 (domestic), any style

PPP-B-636 - Type I or II, class 1, any style, except FTC, TS, OPF and FPF

5.4.3.1 The gross weight for a Level C shipping container shall not exceed 200 pounds for a wood container and the weight limits specified in Specification PPP-B-636 shall not be exceeded for a fiberboard container.

5.4.4 Strapping.- Strapping shall be applied as specified in the applicable container specification.

5.5 Marking.

5.5.1 Interior packages.- Each bag and each box shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package.

5.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. Serial numbers of the periscopes in a Level A or B shipping container shall be added in the identification data and omitted on the Level C shipping container. Fragile markings shall be applied as specified in Standard MIL-STD-129.

5.5.3 Stock number.- The stock number to be added in the identification data shall be obtained from the contracting officer.

5.6 Preproduction pack.- Prior to quantity production, preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

5.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging, and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated.

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6. NOTES

6.1 Intended use.- This periscope is an observation and sighting instrument, designed for direct fire control of certain tank guns. This periscope differs from the normal wherein the gun laying reticle with separate boresight mechanisms are optically projected into separate and independently collimated optical systems by the Cant Corrector M3 for the 8 power optical system and the Infinity Sight M4 for the unity power system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Size and weight of packed shipping container.- The following inside dimensions, overall cubage and gross weight are approximate for six unit packaged periscopes packed in a nailed wood box for Level A:

Inside dimensions -	31. inches long by 25. inches wide by 22½ inches deep
Cubage	10. cubic feet
Gross weight	200. pounds

6.5 Definitions.

6.5.1 Gage.- A gage includes all precision standards and applicable accessories which are calibrated to one basic size to provide an accurate reference dimension.

6.5.2 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

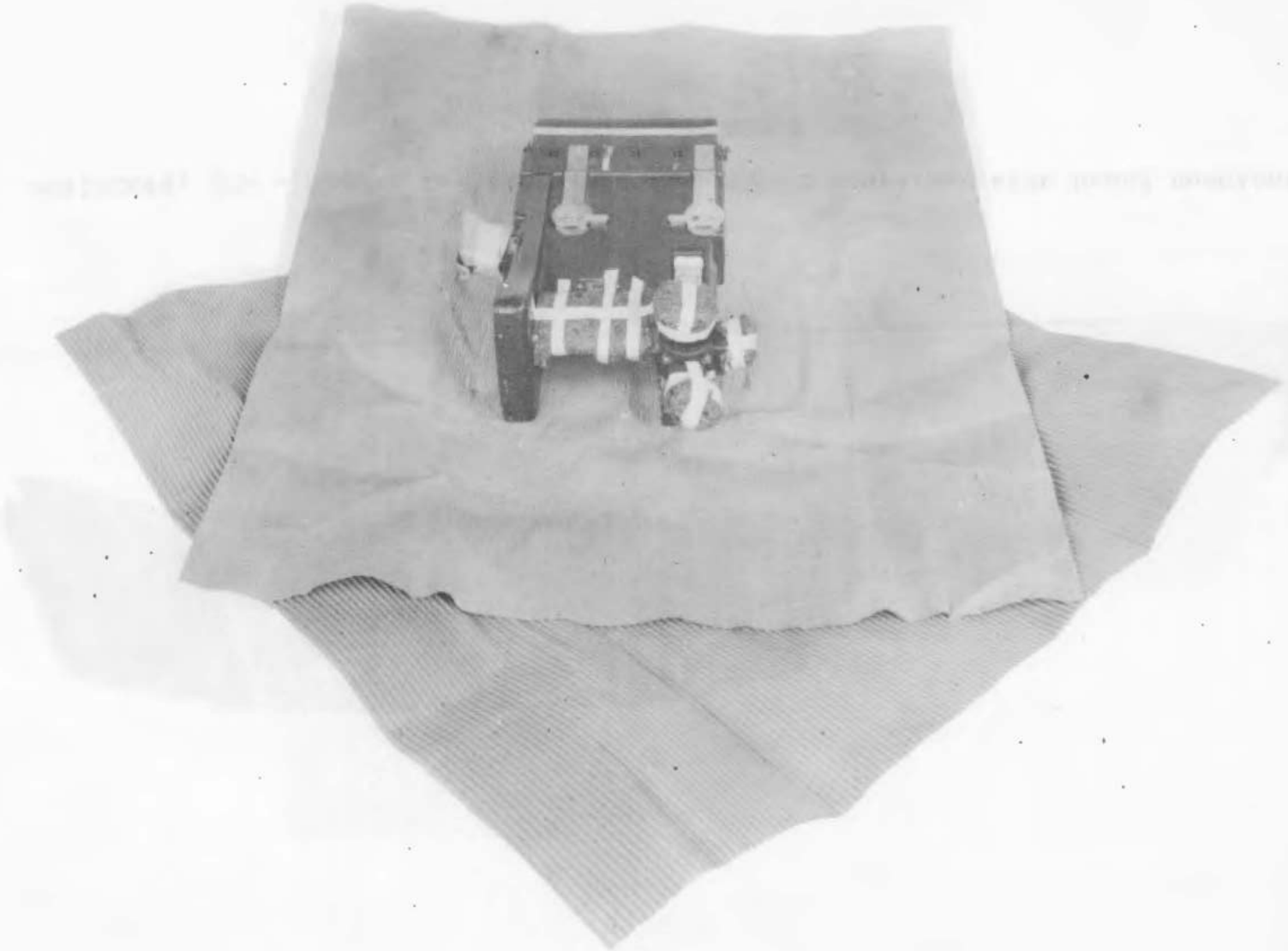
6.5.3 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, and so forth.

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PERISCOPE, M29 - Periscope prepackaged showing optics cushioned.

FIGURE 1

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PERISCOPE, M29 - Bagged Periscope and fiberboard container with bound cushioning.

FIGURE 2

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PERISCOPE, M29 - Packaged periscope.

FIGURE 3

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